Train a Regression Model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Train a Regression Model before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Train a Regression Model trace and record the first step you can explain without help.

Use W2: Score validation for **Fitting** a line. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Train a Regression Model explanation against the supplied facts before discussing it.

Use the worked case for **Fitting** a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10), (4, 15) with MAE.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Train a Regression Model target to a trace and a justified conclusion.

Training points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Train a Regression Model case and show the deciding evidence.

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A regression model has validation MAE 2.4 cm; a **training**-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 22 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Train a Regression Model adds to Responsible Plant Classifier and keep the evidence.

Add the regression experiment to the project appendix only as a separate, clearly limited model.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Training points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Trace candidates

> Trace candidates. Apply the supplied Train a Regression Model facts and keep your reasoning.

Give the three **training** SSE values and selected line.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Score validation

> Score validation. Apply the supplied Train a Regression Model facts and keep your reasoning.

Predict at x 3 and x 4 and calculate MAE/MSE.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Compare the baseline

> Compare the baseline. Apply the supplied Train a Regression Model facts and keep your reasoning.

Fit the **training** constant and score it on validation.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Train a Regression Model facts and keep your reasoning.

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10), (4, 15) with MAE.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit extrapolation

> Limit extrapolation. Apply the supplied Train a Regression Model facts and keep your reasoning.

Why is an exact **training** line not proof of correct **prediction**s at x 100?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Train a Regression Model

> Fit a toy growth line from training pairs, then leave evaluation labels out of fitting. Explain the labelled step before checking the changed condition.

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

Training(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/22

Train a Regression Model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Calculate training fit

2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5.

↓

2. Select line

Choose2x+1 under minimum training SSE.

↓

3. Evaluate once

At day3 predict7 versus observed6: absolute error1 cm.

CHANGE ONE THING

Evaluation observation becomes8 instead of6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At day3 predict7 versus observed6: absolute error1 cm.

1. Select line

2. Calculate **training** fit

3. Evaluate once

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Evaluate once At day3 predict7 versus observed6: absolute error1 cm. Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Evaluation **observation** becomes8 instead of6. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Absolute evaluation error stays1 cm, but residual changes from-1 to+1. **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Repair a missing building block

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

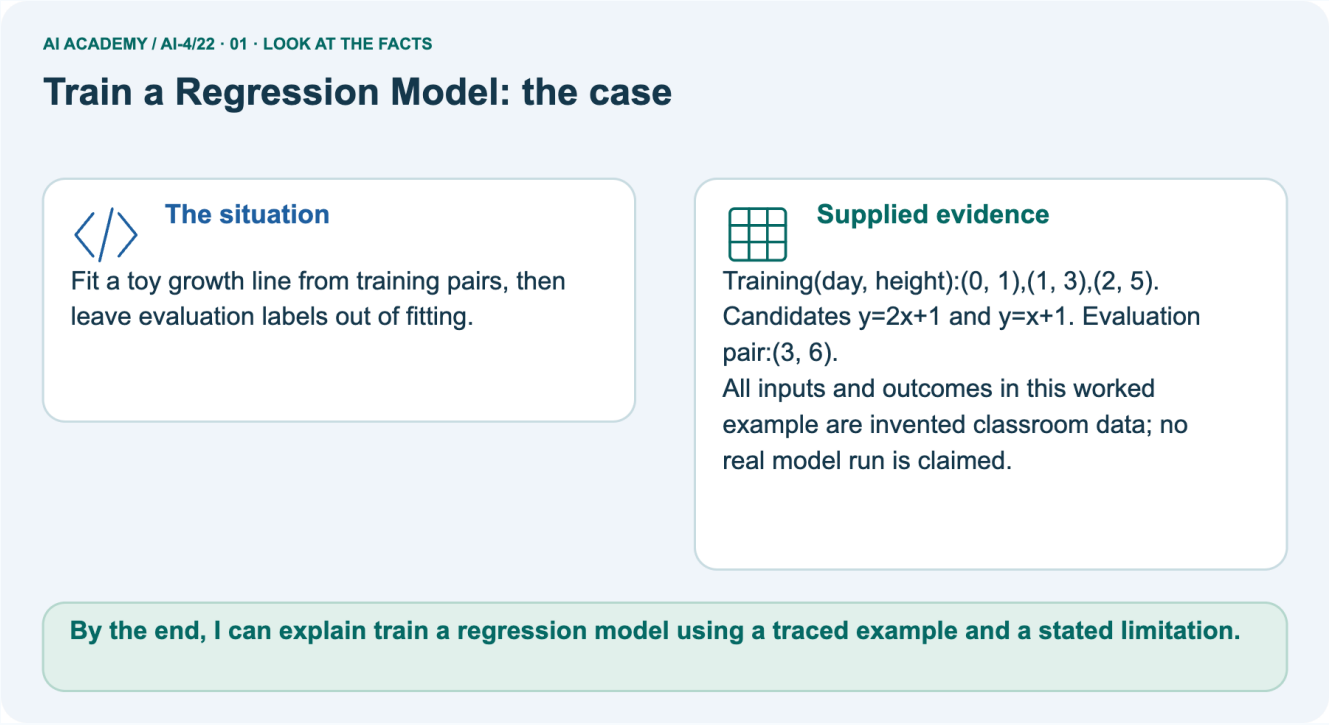
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Train a Regression Model picture from its labels, then record a relevant limit.

01. Train a Regression Model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

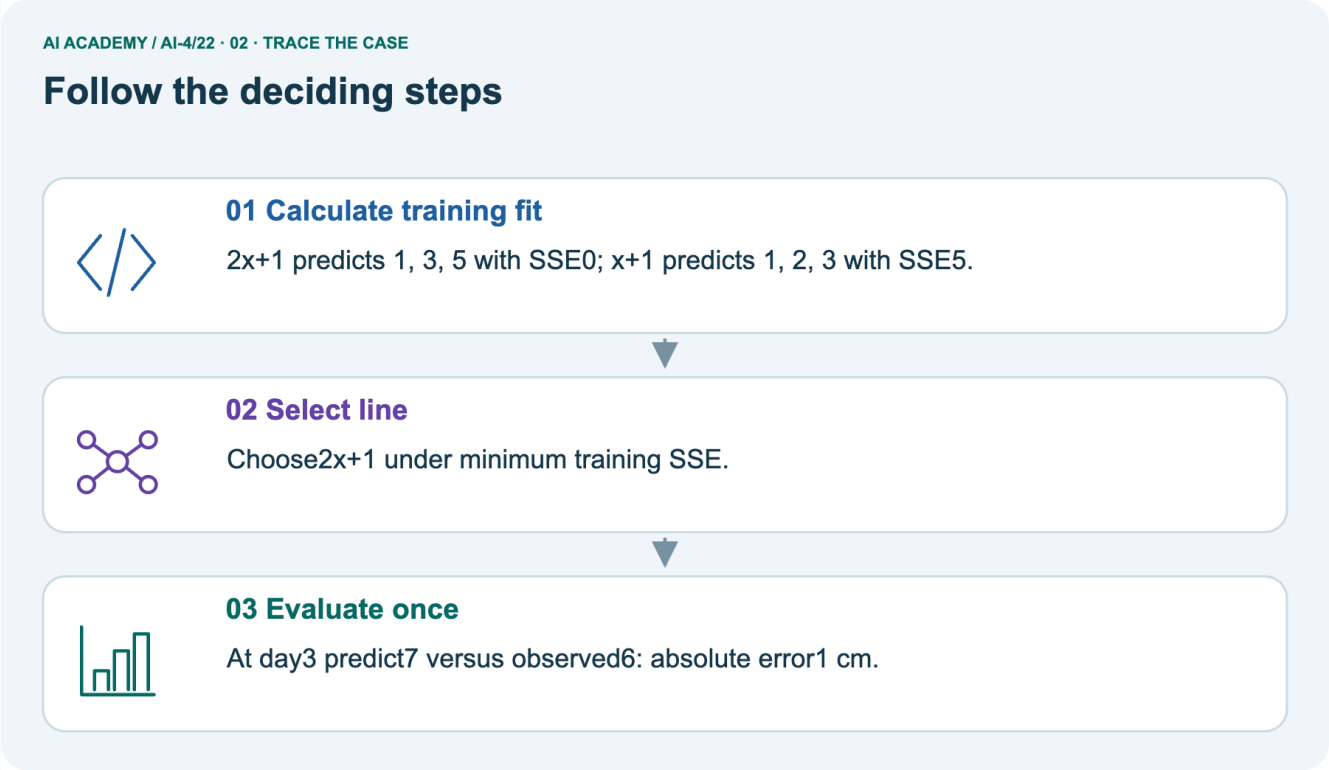
Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**. Supplied facts: **Training**(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



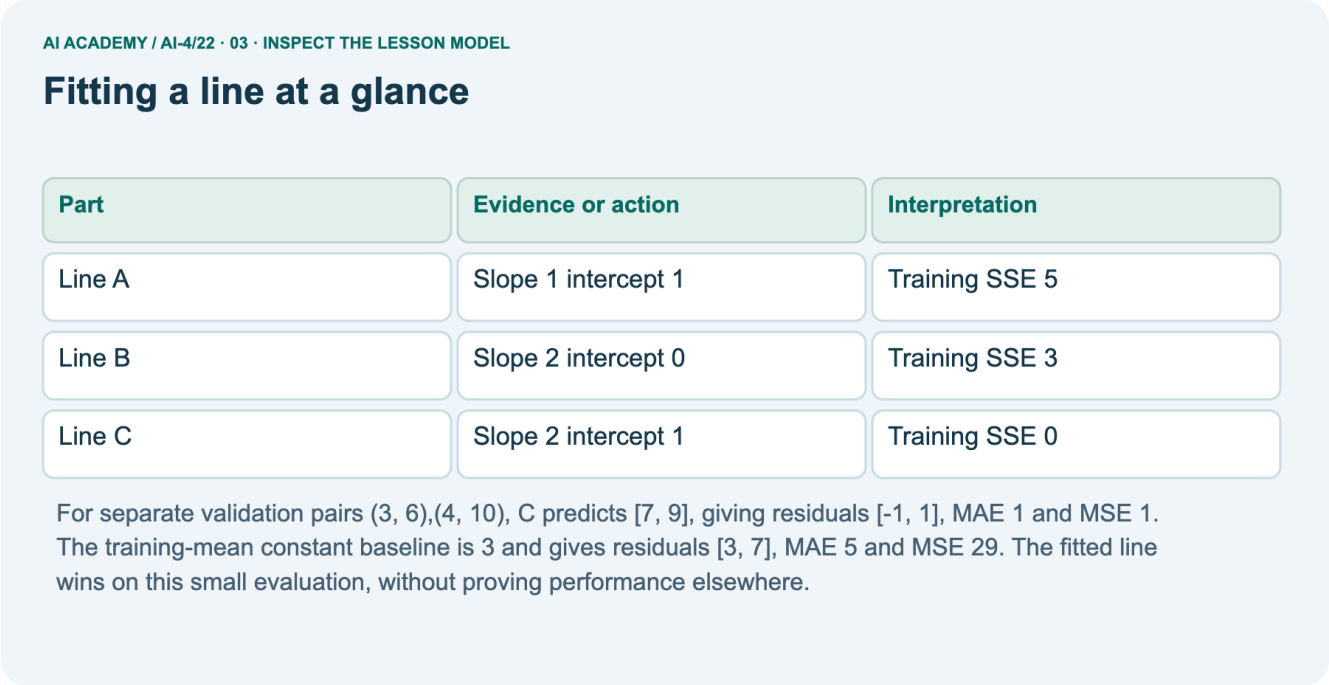
Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **training** fit: 2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5. → Select line: Choose2x+1 under minimum **training** SSE. → Evaluate once: At day3 predict7 versus observed6: absolute error1 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Fitting a line at a glance



AI ACADEMY / AI-4/22 · 03 · INSPECT THE LESSON MODEL

Fitting a line at a glance

Part	Evidence or action	Interpretation
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

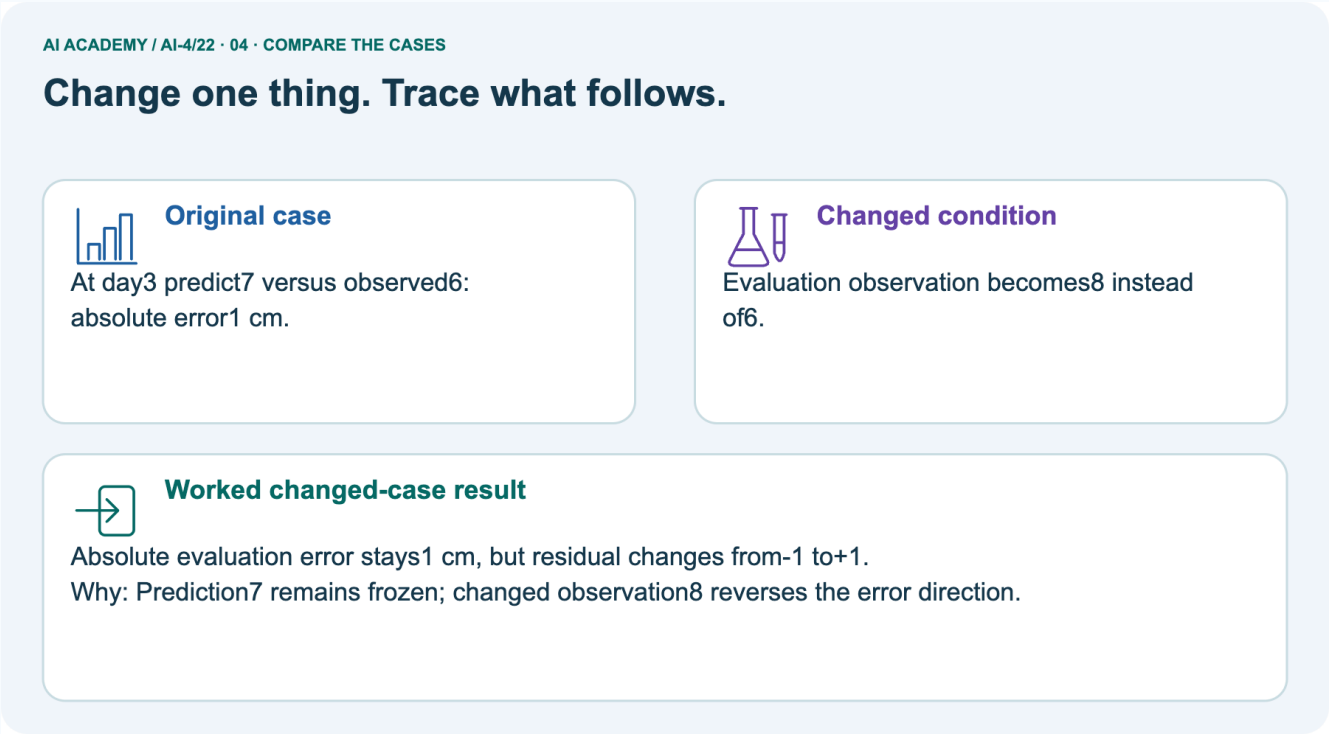
Follow the labelled picture with the matching explanation. Open the vector diagram

Fitting a line at a glance. Line A | **Slope** 1 **intercept** 1 | **Training** SSE 5. Line B | **Slope** 2 **intercept** 0 | **Training** SSE 3. Line C | **Slope** 2 **intercept** 1 | **Training** SSE 0

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

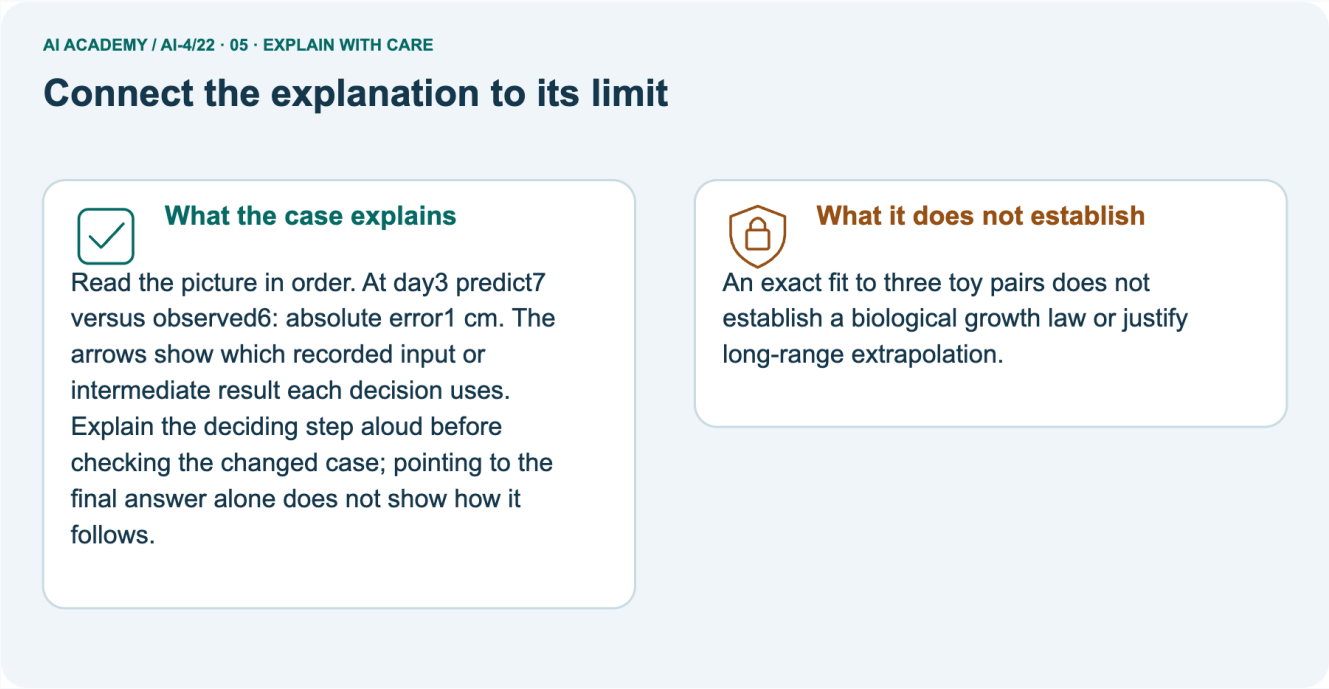
Original case: At day3 predict7 versus observed6: absolute error1 cm. Changed condition: Evaluation **observation** becomes8 instead of6. Expected result: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

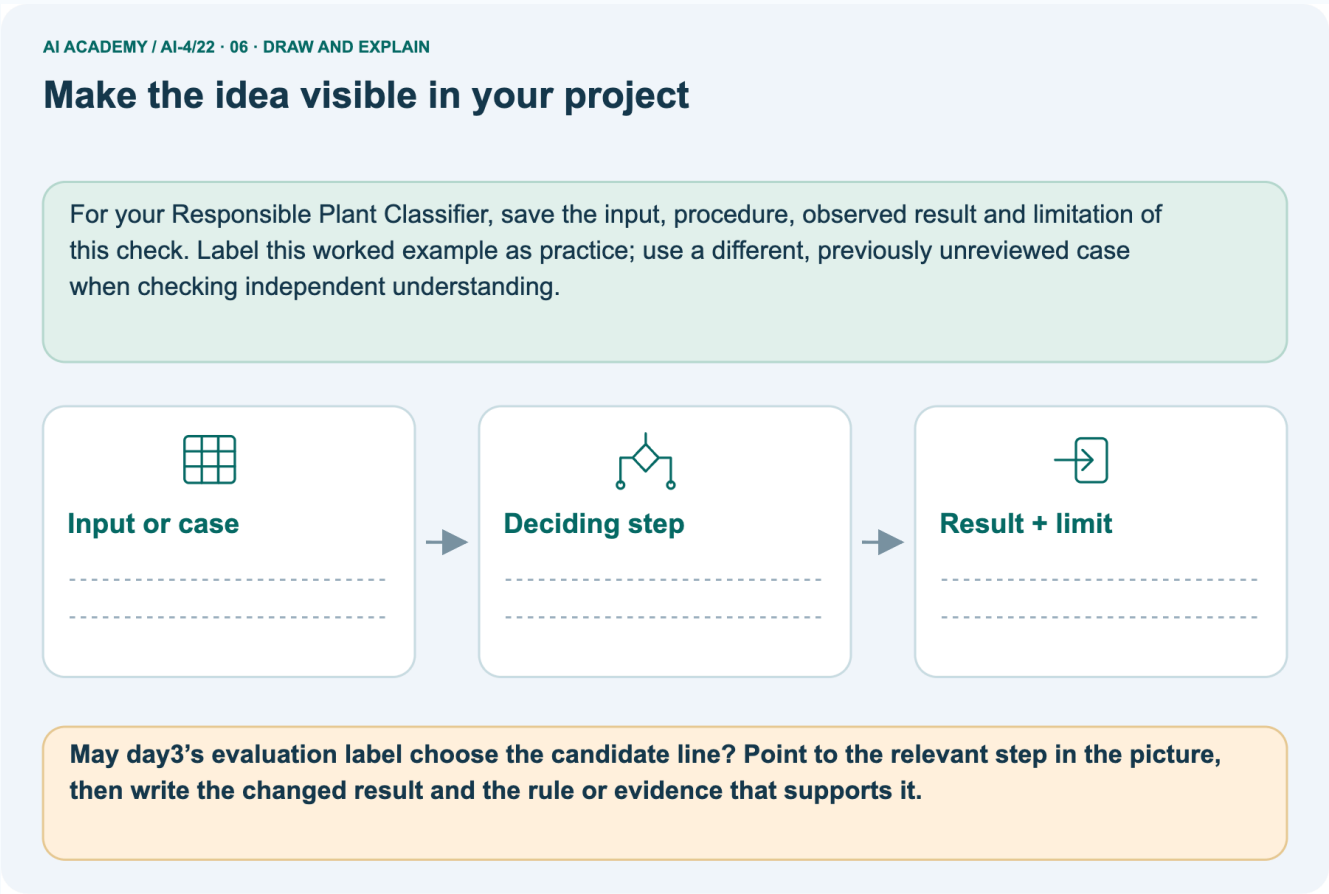
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.